

RENEAL TANZANIA

School Lab Lag Analytics: System Purpose & Documentation

A data-driven information platform for monitoring lab usage patterns, identifying under-performing schools and guiding NECTA ICT exam readiness across 115 government secondary schools in 11 regions of Tanzania data up to 2025.

Given all other factors, the Schools should have to learn from the Neighboring Reneal schools Lab that perform ICT National Examination as the known information, but also how to improve the usage if asking the teacher concerned. 28 Schools have poor-usage while the nearby schools have good-usage while 37 schools have not done ICT Necta and have required usage and while the nearby schools that have done so. But also the ones with low and moderate usage and still have done ICT Necta exams. Hence this system points out the schools even for more recognition.

Core Metrics Available

- **Schools Monitored:** 115 across 11 regions
- **Data Years:** 2019 – 2025
- **Poor Usage Schools:** 28 (Need intervention)
- **No NECTA + High Usage:** 37 (Pending to sit for exam)
- **Poor Usage & Moderate + NECTA:** 19

Summary

The Reneal Tanzania School Lab Analytics is a self-contained, browser-based analytical application purpose-built to help the Reneal Tanzania NGO and its partners make data-informed decisions about LTSP computer lab performance across Tanzania's government secondary schools network.

Since 2013, Reneal Tanzania has partnered with Tanzanian districts to install and maintain computer labs in government secondary schools. By 2026, this network spans 115 active schools across 11 regions. While the physical infrastructure is in place, the organization faces two persistent and interrelated challenges:

- **01. Uneven Lab Utilization:** A subset of schools records critically low student hours despite having functioning equipment, trained teachers, and proximity to high-performing neighboring schools.

- **02. Missed NECTA ICT Exam Opportunity:** Dozens of schools with strong lab usage records have never registered students for the national ICT Computer Studies examination, leaving students without a formal credential for their digital skills.

The Page directly addresses both challenges. It uses three primary data sources—GPS coordinates, lab usage logs (2019–2025), and NECTA ICT exam results (2018–2025)—cross-references them using distance neighbor matching and presents actionable intelligence across three analytical views: Usage Checkup, NECTA Checkup, and an Overview.

The application is entirely self-contained that runs in any modern web browser and supports PDF export of all tabular data for reporting and sharing with various officials.

Organizational Context & Purpose

The programme now covers 115 active schools till early 2026 across the following 11 regions: Arusha, Kilimanjaro, Manyara, Dodoma, Iringa, Pwani, Mara, Tanga, Mbeya, Dar es salaam and Lindi. Each school is assigned to a technical team member responsible for maintenance and capacity building.

The Problem the Dashboard Solves

As the network has grown two monitoring challenges have become increasingly visible to the Reneal leadership team:

- **Under-utilized Labs:** Some schools log very few student hours per year—sometimes under 100 hours—while schools nearby in the same region or district record tens of thousands of hours. Without a systematic view of this disparity, it is difficult to know where to direct field visits, mentoring resources, or ICT teacher training.
- **Unregistered NECTA ICT Candidates:** Tanzania's National Examinations Council (NECTA) offers a Computer Studies examination as part of the CSEE (Certificate of Secondary Education Examination). Schools with operational labs and strong usage records should be registering Form IV students for this exam. Yet a large proportion of Reneal schools—including many with good usage—have never done so, representing a missed opportunity for students to earn a nationally recognized ICT credential.

The Dashboard provides a single interface that surfaces both problems simultaneously, ranked by severity, enriched with geographic neighbor data, and exportable for use in district meetings and donor reports.

Purpose

The system serves four strategic purposes for Reneal Tanzania:

- **Accountability:** Enabling leadership to see at a glance which schools are performing poorly and which are thriving, without requiring manual spreadsheet analysis.
- **Peer Learning Facilitation:** By computing the nearest high-performing school for every under-performing school, the dashboard identifies natural mentoring pairs that field teams can activate through cross-school visits.
- **NECTA Advocacy:** By surfacing schools that have strong usage but no exam history, the dashboard provides evidence for conversations with school heads and district education officers about registration.
- **Partner Reporting:** The PDF export feature allows Reneal to include current-filter snapshots in grant reports, partner updates, and district coordination meetings.

Methodology

WHY NEIGHBORS MATTER: Geographic proximity is the most actionable form of peer comparison for Reneal's field model. A school 8km away from a high performer is a realistic candidate for a cross-school visit, a shared ICT teacher workshop, or a management exchange. Distance data turns abstract rankings into field-ready action plans.

Usage Level Classification

The recent average is the mean of the last three calendar years for which non-zero data exists. For example, a school that only has data for 2024 and 2025 is averaged over those two years only.

NECTA Readiness Classification

For schools that have not yet sat the ICT exam, the Dashboard assigns a readiness tier based on average recent usage:

- **Ready:** Usage $\geq$ 2,000 hours per year. Students have had sufficient computer time to prepare for the exam.

- **Preparing:** Usage $\geq 1,000$ and $< 2,000$ hours per year. The lab is active and building toward exam readiness.
- **Early:** Usage $< 1,000$ hours per year. The lab is not yet active enough to support exam preparation.

Application & UI Structure

Navigation Bar

A sticky header bar sits at the top of every page to facilitate smooth transitions between analytical modules:

- **Tab 1: Usage Checkup**
 - *Function:* Navigates to the lab usage analysis view (marked with a red indicator dot).
 - *Purpose:* This is the primary intervention view. It displays all 115 schools in a sortable, filterable table ranked by average usage, identifying under-performing schools and their nearest high-performing neighbors.
- **Tab 2: NECTA Checkup**
 - *Function:* Navigates to the NECTA exam participation view (marked with an indigo indicator dot).
 - *Purpose:* This view focuses on the NECTA ICT exam opportunity gap highlighting schools that have good or moderate lab usage but have never registered students for the national exam.
- **Tab 3: Overview**
 - *Function:* Navigates to system-wide summary charts and top-20 rankings (marked with a teal dot).
 - *Purpose:* Provides a system-wide analytical perspective using four visualizations, optimized for leadership reviews, presentations and coordination meetings.
- **Pill Badges:** Located on the far right of the nav bar, these small badges show live summary counts of poor-usage schools and "good usage + no NECTA" schools updating automatically as data loads.

Stat Strip (KPI Cards)

Each analytical tab opens with a row of four summary KPI cards providing an at-a-glance count before digging into the tables. The bottom edge of each card is color-coded by significance.

- **Poor Usage Tab Cards:** Poor Usage (count $\leq 500h$) | Moderate (500–2000h) | Good Usage ($\geq 2000h$) | No 2025 Data.
- **No NECTA Tab Cards:** No NECTA + Good Usage | No NECTA + Moderate | Has

PDF Export Capabilities

Both primary data tables support direct PDF export. Two export buttons sit in the toolbar row to the right of the search input:

- **Current view as PDF:** Exports exactly what is currently visible in the table—respecting active filters, sorting directions and search queries. The PDF header labels the active filter and school count.
- **All schools as PDF:** Exports the complete 115-school dataset regardless of onscreen filters, sorted by usage ascending (Tab 1) or descending (Tab 2) for comprehensive reporting.

Both exports generate an **A4 landscape PDF** including a title header, a subtitle with filter status and generation date, a color-coded legend strip (using filled squares for font compatibility), alternating row shading, and consistent page numbering. Files are named automatically: reneal-[report-type]-[DD-Mon-YYYY].pdf.

Key Findings from the Data (As of May 2026)

Poor Usage — Critical Cases

Fourteen schools (28) are currently classified as Poor Usage (< 500h average).

Notable patterns include:

- **Extreme Disparities:** The gap between the worst-performing school and its nearest high-usage neighbor is often dramatic. For example, one school logging fewer than 10 hours per year sits within 5km of a school logging over 27,000 hours per year.
- **Geographic Feasibility:** All 28 poor-usage schools have at least one good-usage school within 100km, making cross-school peer learning logistically viable.
- **Operational Overlap:** Several poor-usage schools are in the same district and assigned to the same technical team member as high-usage schools, suggesting that targeted training could yield rapid improvements.

NECTA Participation Gap

Of the 115 schools, 76 have never sat the ICT NECTA exam. Among these:

- **Missed Credentials:** 37 schools exhibit Good Usage (≥ 2000 h average). Students in these schools have extensive lab time but graduate without a formal ICT credential.
- **Poor and Moderate:** 19 schools that have done NECTA despite Poor and Moderate student usage hours.
- **Proximity to Help:** The nearest NECTA participating school for non-participating

institutions is, in most cases, within 50km—well within regional coordination distances.

- **High-Value Targets:** Several of the highest-usage schools in the entire network (over 15,000 annual hours) have never registered for the exam, representing the largest single opportunity for credentialing impact.

Regional Patterns

- **High Performance Tiers:** Arusha and Kilimanjaro regions have the highest density of good-usage schools, reflecting earlier programme rollout and stronger ICT teacher capacity.
- **Newer Deployments:** Newer regions (Lindi, Iringa, Mara) have a higher proportion of moderate and poor-usage schools, consistent with shorter programme tenure.
- **Systemic Barriers:** NECTA participation is heavily concentrated in specific regions; several regions have no participating schools at all.

User Manual & Workflows

Workflow

1. Open the **Poor Usage** Check up tab and click the "**Poor <500h**" filter to isolate critical schools.
2. Click an individual poor-usage school row to expand its details; note its nearest high-usage neighbor and distance.
3. Click an individual Poor + Has NECTA and Moderate + Has NECTA and observe the schools.
4. Click "**Download Current view as PDF**" to export the targeted list for district meeting agendas.
5. Switch to the **NECTA** Checkup tab and click "**No NECTA + Good**" to see schools most ready to register for national exams.
6. Note the nearest NECTA participating school to identify natural partners for registration guidance.

Workflow for Overview

1. Open the **Overview** tab for a system-wide strategic perspective.
2. Review the **Year-over-Year trend chart** to assess overall network growth trajectories.
3. Use the **NECTA vs No NECTA chart** to quantify the exam participation gap for donor presentations.

4. Use the **Regional breakdown** to prioritize field resource allocation for upcoming maintenance cycles.
5. Export "**All schools as PDF**" from Tab 1 or Tab 2 for clean, comprehensive annual reporting.

Sorting and Searching

Every column header with a  arrow is fully sortable. Click once to sort ascending, click again to sort descending. The search box filters school names, districts, and regions simultaneously, allowing you to pull up specific records instantaneously during field calls.

Adjusted under-performing 500hrs to 1000hrs

14 schools reclassified from Moderate → Poor (all had avg < 1000h but were carrying the old ul='moderate' label from when the threshold was 500):

School	Avg hrs
Mringa	988h
Bagara	961h
Manyara Girls	942h
Butuguri	866h
Ilkiding'a	842h
Kisaza	822h
Nkoasenga	814h
Ngiresi	781h
Suye	720h
Engarenaibor	691h
Oldadai	614h
Mukulat	598h
Kwaraa	585h
Sing'isi	556h

Final counts: Poor = 28 schools · Moderate = 30 · Good = 57 · Total = 115. The ul field in neighbour (ng) references was also corrected where affected.

School Size Comparison — Explanation

The School Size Comparison tab answers one core question: **does the size of a school's student population shape how it performs on the NECTA ICT exam?** It does this by cross-referencing two variables — enrollment figures from Reneal's school records and ICT exam results from the national NECTA database — and putting them into a single analytical frame covering 133 exam records across 51 schools, 8 years (2018–2025), and both Form 2 and Form 4.

The central unit is the **exam record**: one school, one year, one form level. Each record carries enrollment at that point in time, the number of students who actually sat the ICT exam, the breakdown of grades awarded (A through F), and the overall institution grade assigned by NECTA. That combination lets you compute two things the raw results spreadsheet never directly tells you: **participation rate** (how many enrolled students actually sat the exam) and **grade quality** (what fraction of those who sat earned A or B versus failed outright).

The Five Size Tiers

Schools are grouped into five enrollment bands:

Tier	Enrollment Range	Records	Avg Enrollment
XS	Under 150	21	~104
SM	150–299	54	~230
MD	300–449	44	~357
LG	450–599	13	~504
XL	600+	1	715

The bands are not arbitrary. They roughly correspond to operational thresholds that matter for a rural secondary school in Tanzania: XS schools are typically single-stream with one Form 2 and one Form 4 class, SM schools are the most common configuration in Reneal's network, MD and LG schools are multi-stream institutions that have grown significantly, and the single XL entry — Longido A-Level SS in 2025 — is a genuine outlier, a large A-Level institution that sits Form 2 alongside its upper-secondary cohorts.

Sample of Data Collected

#	School	A	B	C	D	F	Sat for Exam	Grade	Total students in the classroom
1	Enyolito	0	0	5	9	57	71	Grade F (Fail)	252
2	Engamaibor	0	1	8	3	8	20	Grade D (Satisfactory)	260
3	Arusha Day Sec	0	0	1	4	8	13	Grade D (Satisfactory)	263
4	Mateves	0	5	14	2	0	21	Grade C (Good)	153
5	Longido Sec	0	0	8	22	26	56	Grade D (Satisfactory)	519
6	Miangarini	0	1	4	12	14	31	Grade D (Satisfactory)	224
7	Tingatinga Sec	0	5	20	11	0	36	Grade C (Good)	312
8	Suye Sec	0	0	3	10	30	43	Grade F (Fail)	330
9	Mwandet	0	1	20	14	1	36	Grade C (Good)	395
10	Osiliqi	0	0	4	12	1	17	Grade D (Satisfactory)	111
Total		0	13	87	99	145	344		

ICT NECTA RESULTS 2023 – Form 2									
#	School	A	B	C	D	F	Sat for Exam	Grade	Total students in the classroom
1	Enyolito	0	0	3	12	9	24	Grade D (Satisfactory)	198
2	Engamaibor	0	1	13	3	1	18	Grade C (Good)	326
3	Arusha Day Sec	1	1	12	7	3	24	Grade C (Good)	287
4	Longido Sec	0	0	21	13	1	35	Grade C (Good)	589
5	Miangarini	0	0	6	12	4	22	Grade D (Satisfactory)	287
6	Tingatinga Sec	0	3	26	10	2	41	Grade C (Good)	373
7	Suye Sec	0	1	8	16	5	30	Grade D (Satisfactory)	236
8	Mwandet	1	7	13	4	0	25	Grade C (Good)	532
9	Osiliqi	0	2	10	9	0	21	Grade C (Good)	104
Total		2	15	112	86	25	240		

ICT NECTA RESULTS 2024 – Form 2									
#	School	A	B	C	D	F	Sat for Exam	Grade	Total students in the classroom
1	Enyolito	0	2	20	14	2	38	Grade C (Good)	177
2	Engamaibor	0	1	5	9	4	19	Grade D (Satisfactory)	357
3	Arusha Day Sec	0	4	23	6	0	33	Grade C (Good)	312

Data Sources

The Dashboard is powered by three datasets merged and embedded directly into the application.

1. GPS Coordinates

- **Source file:** 2026-03-24-gps-coordinates.csv
- **Description:** This dataset provides the geographic anchor for every school in the network. Each record contains the school's full name, district, and precise latitude/longitude coordinates obtained from field GPS measurements.
- **Application:** These coordinates drive the geographic scatter visualization in the Overview tab and are used to compute inter-school distances for neighbor matching.

NOTE: Coordinate data covers 115 active schools. Schools with missing or implausible coordinates are excluded from neighbor matching calculations to prevent geographically erroneous pairings.

2. Lab Usage Data (2019–2025)

- **Source file:** school-usage-2019-to-2025.xlsx
- **Description:** This primary performance dataset contains annual student hours

logged by each school's LTSP server from 2019 through 2025. The data is sourced from Reneal's reporter data collection.

- **Key Fields:** School name/identifier, district/region assignments, assigned technical team member, annual usage figures (2019–2025) and the 2025 national usage rank.

The dashboard computes a **recent average** metric for each school—the mean of the last three years for which data is available. This smooths out anomaly years (e.g., COVID-related closures in 2020–2021) and gives a more stable performance signal. Schools are classified into three tiers based on annual student hours:

Tier	Recent Average Usage Threshold	Status Description
Good Usage	$\geq 2,000$ hours / year	Strong student engagement
Moderate Usage	500 – 2,000 hours / year	Growing but needs support
Poor Usage	< 500 hours / year	Critical intervention needed

3. NECTA ICT Exam Results (2018–2025)

- **Source file:** Results-Analysis-2018-2025.xlsx
- **Description:** This dataset records ICT Computer Studies examination results from NECTA for Form II and Form IV students across Reneal supported schools. Each record captures the year, form level, school name, candidate count, and grade distribution (A through F).

The Dashboard uses this data to answer quantitative question per school:

1. Has the school ever participated in an ICT NECTA exam?

Schools that have never appeared in the NECTA results dataset are flagged as non-participants. When combined with a "Good Usage" tier, this produces the **"No NECTA + Good Usage"** classification that drives the second analytical view.

